Mit Soneshbhai Patel

Portfolio | LinkedIn | GitHub
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Summary

Final-year Computer Science student specializing in Machine Learning, Deep Learning, NLP, and Generative AI. Builds AI-powered systems and scalable APIs with LLMs, LangChain, RAG, FastAPI, TensorFlow, and Hugging Face. Applies end-to-end ML pipelines spanning data preprocessing, model training, evaluation, deployment, and inference.

Education

Pandit Deendayal Energy University 2023 – 2027
Bachelor of Technology, Computer Science & Engineering *CGPA: 8.97/10*

Experience

Astha Technology Solutions Pvt. Ltd. May 2026 – Jul 2026
Machine Learning Engineer Intern *Surat, India*

- Engineered an LLM-powered recruitment platform using Anthropic Claude, local-language models, Django, provider fallbacks, and structured outputs to automate job descriptions, resume screening, and candidate evaluation.
- Built personalized interview questions by analyzing resumes and job descriptions, detecting seniority, and generating tailored HR, technical, situational, and experience-based questions with expected-answer guidance.
- Built AI-powered Django REST APIs with PostgreSQL, LLMs, and background threads to automate recruitment.
- Built financial KPIs, administrative dashboards, and departmental insights via Django ORM and SQL.

Projects

US Visa Approval Prediction System | *Live* | *Code* May 2026

- Developed a modular end-to-end machine learning pipeline for 25,480 US visa applications, implementing data validation, preprocessing, feature engineering, XGBoost training, evaluation, and thresholded real-time inference.
- Optimized an XGBoost classifier using Optuna hyperparameter tuning and a 0.45 decision threshold; MLflow logged 91.83% recall, 73.07% precision, 81.39% F1-score, and 71.94% accuracy on 5,096 held-out applications.
- Built Dockerized FastAPI app for real-time inference using shared preprocessing and artifacts.

Deepfake Video Detection System | *Code* Jun 2026

- Developed a deepfake detection app using TensorFlow/Keras, OpenCV, Flask, and Gunicorn; achieved 0.847 ROC-AUC, 0.95 recall, 0.921 F1-score, and 0.738 balanced accuracy on 99 DFDC test videos at a 0.5 threshold.
- Engineered a face-focused pipeline with YuNet detection, InceptionV3 transfer learning, and a mask-aware two-layer GRU, raising ROC-AUC from 0.59 to 0.847 through face-cropped 20-frame temporal sequence analysis.
- Built Flask inference for video uploads with REAL/FAKE labels, confidence, and class probabilities.

Movie Recommendation System | *Live* | *Code* Jun 2025

- Built a content-based movie recommender for 4,806 TMDB titles by merging movie and credit datasets, parsing JSON metadata, and engineering NLP tag profiles from overviews, genres, keywords, cast, and directors.
- Applied NLTK Porter stemming and CountVectorizer (5,000 features) to generate a 4,806 x 4,806 cosine-similarity matrix from overview, genre, keyword, cast, and director tags, ranking top five recommendations per title.
- Built a Streamlit app with searchable titles, top-5 recommendations, and TMDB API poster retrieval.

Technical Skills

Programming Languages: Python, SQL, C
Core Areas: Machine Learning, Deep Learning, Natural Language Processing, Generative AI
Frameworks & Libraries: TensorFlow, LangChain, RAG, Scikit-Learn, FastAPI, Hugging Face, Pydantic
Tools & Platforms: Git, GitHub, Docker, Streamlit, Linux/Bash
Databases: MySQL, PostgreSQL, MongoDB

Certifications

Google Agentic AI Hackathon | View Certificate
Microsoft Power BI | View Certificate